

PROFESSIONAL INSTALLATION MANUAL

Ubuntu Server Deployment and Initial Configuration

ITEP 414 – System Administration and Maintenance

Mini Project – Professional Installation Manual

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I. Introduction

Ubuntu Server is a Linux-based operating system designed for servers and other computing environments that require reliable and efficient system services. It is commonly used for web hosting, file sharing, database management, application hosting, networking services, and other server-related tasks. Because Ubuntu Server does not require a graphical desktop environment by default, it can operate efficiently using fewer system resources and can be managed primarily through the command line.

This installation manual provides a complete procedure for installing and configuring Ubuntu Server in a virtual machine environment. The purpose of this guide is to provide a clear and organized procedure that another system administrator can follow without requiring additional instructions. The manual covers the installation process, initial configuration, verification procedures, troubleshooting techniques, and recommended administrative practices.

Before beginning the installation, the administrator should ensure that the required hardware or virtual machine resources are available. A stable internet connection is also recommended because software updates and additional packages may need to be downloaded during or after installation.

II. Ubuntu Server Installation Guide

2.1 Prepare the Installation Environment

The first step is to prepare the environment where Ubuntu Server will be installed. For a virtual machine installation, create a new virtual machine using virtualization software such as VirtualBox or VMware. Assign appropriate resources to the virtual machine. For a basic server installation, a minimum of 2 GB of RAM, at least 2 CPU cores, and approximately 25 GB of virtual disk storage may be used. Additional resources can be assigned depending on the services that the server will provide.

The Ubuntu Server installation ISO should be downloaded from the official Ubuntu website. After downloading the ISO file, configure the virtual machine to use the ISO as its bootable optical disk. If installing Ubuntu Server on physical hardware, create a bootable USB installer and configure the computer to boot from the USB device.

2.2 Start the Installation

Start the virtual machine or physical computer and allow it to boot from the Ubuntu Server installation media. The Ubuntu Server installer will display the initial installation screen. Select the appropriate language and continue to the next step.

The installer will then ask for the keyboard layout. Select the keyboard layout that matches the physical keyboard being used. The layout can usually be detected automatically, but it should still be checked to ensure that the correct configuration is selected.

2.3 Configure the Network

During installation, Ubuntu Server will attempt to detect the available network connection. If the system is connected to a network, the installer should automatically obtain

an IP address through DHCP. Verify that the correct network interface has been detected and that the connection is functioning properly.

For a basic installation, the default DHCP configuration can be used. A static IP address can be configured later if the server will provide services that require a permanent network address.

2.4 Configure Storage

The installer will provide several storage configuration options. For a basic virtual machine installation, the guided storage configuration can be selected. This allows Ubuntu Server to automatically create the required partitions and file systems.

Select the virtual disk that will be used for the installation and review the proposed storage configuration carefully. If the disk contains important information, do not continue without confirming that the correct disk has been selected. Once the storage configuration has been reviewed, confirm the changes so that the installer can create the required partitions.

2.5 Create the User Account

The installer will request information for the first administrative user. Enter the administrator's real name, server name, username, and a strong password. The username should be simple and appropriate for system administration.

The password should contain a combination of uppercase and lowercase letters, numbers, and special characters. Avoid using easily guessed information such as names, birthdays, or common words.

2.6 Configure SSH

Ubuntu Server may provide an option to install the OpenSSH server during installation. SSH allows administrators to remotely access and manage the server through a secure command-line connection.

If the server will be administered remotely, select the option to install OpenSSH Server. If SSH is not required at this stage, it can be installed later using the appropriate package management command.

2.7 Select Additional Software

The installer may provide a list of additional server software or packages that can be installed. Only install software that is necessary for the intended purpose of the server. Installing unnecessary services can consume system resources and increase the server's attack surface.

After selecting the required options, continue with the installation. Ubuntu Server will copy the required files to the disk and configure the operating system.

2.8 Restart the Server

After the installation process has finished, the installer will request that the system be restarted. Restart the virtual machine or physical server and remove the installation ISO or USB installer when prompted. This ensures that the system boots from the newly installed Ubuntu Server operating system rather than returning to the installation program.

After restarting, log in using the administrator account created during installation. The successful login confirms that the basic installation has been completed.

III. Initial Server Configuration

3.1 Update the System

After logging in for the first time, the server should be updated before additional configuration is performed. Updating ensures that the system receives the latest available software packages and security fixes.

```
sudo apt update
sudo apt upgrade -y
```

3.2 Configure the Hostname

A meaningful hostname makes the server easier to identify when managing multiple systems. The hostname should describe the server's purpose, such as web-server, file-server, or ubuntu-server. The hostname can be checked and changed using the following commands.

```
hostnamectl
sudo hostnamectl set-hostname ubuntu-server
```

3.3 Configure Network Settings

The network configuration should be reviewed to ensure that the server has a working connection. The following commands can be used to display network interfaces and test connectivity.

```
ip address
ping -c 4 8.8.8.8
ping -c 4 google.com
```

3.4 Configure SSH

If SSH was not installed during the Ubuntu installation, it can be installed using the following command. The SSH service can then be checked, started, and enabled to start automatically at boot.

```
sudo apt install openssh-server -y
sudo systemctl status ssh
sudo systemctl start ssh
sudo systemctl enable ssh
```

3.5 Configure the Firewall

Ubuntu Server can use UFW, or Uncomplicated Firewall, to manage basic firewall rules. The firewall should be enabled after allowing required services. For a server that uses SSH, allow SSH before enabling the firewall.

```
sudo ufw allow ssh
sudo ufw enable
sudo ufw status
```

3.6 Create Additional Administrative Users

For a production environment, system administrators should avoid sharing a single account. Individual user accounts allow administrators to be identified and managed separately. A new user can be created and granted administrative privileges using the following commands.

```
sudo adduser username  
sudo usermod -aG sudo username
```

IV. Server Verification Procedures

After completing the installation and initial configuration, the administrator should verify that the server is operating correctly. Verification helps identify configuration problems before the server is placed into production.

The operating system version, kernel version, hostname, network configuration, connectivity, services, firewall, disk space, memory usage, and system logs should be reviewed using the following commands.

```
lsb_release -a  
uname -r  
hostnamectl  
ip address  
ping -c 4 8.8.8.8  
ping -c 4 google.com  
systemctl status ssh  
sudo ufw status  
df -h  
free -h  
sudo journalctl -p err -b
```

A server can be considered successfully configured when the operating system boots normally, the administrator can log in, the network connection works, required services are running, the firewall is active, and there are no unresolved critical system errors.

V. Troubleshooting Guide

5.1 Server Does Not Boot

If Ubuntu Server does not boot after installation, first verify that the installation media has been removed or that the virtual machine is configured to boot from the virtual disk. Check the virtual machine's boot order and confirm that the correct disk is selected. If the problem continues, review the installation process and verify that the operating system was successfully installed on the intended disk.

5.2 No Internet Connection

If the server cannot access the internet, check the network interface using the command below. Confirm that the interface has an IP address. Test connectivity to the network gateway and then test an external IP address. If an IP address is available but domain names cannot be resolved, investigate the DNS configuration.

```
ip address
```

5.3 SSH Connection Fails

If a remote SSH connection cannot be established, verify that the SSH service is running. Check that SSH is listening on the expected port and confirm that the firewall allows SSH traffic. The server's IP address should also be verified to ensure that the administrator is connecting to the correct system.

```
sudo systemctl status ssh
```

5.4 Firewall Blocks a Required Service

If a service cannot be accessed because of the firewall, check the current firewall rules. If the service requires a specific port, allow only that port. For example, a web server using HTTP may require port 80.

```
sudo ufw status
```

```
sudo ufw allow 80/tcp
```

5.5 Insufficient Disk Space

If the server reports that the disk is full, use the `df -h` command to identify available storage. Large files and unnecessary packages should be identified and removed carefully. System logs should also be reviewed if they are consuming excessive storage. Important files should never be deleted without confirming their purpose.

```
df -h
```

5.6 Service Is Not Running

If an installed service is not functioning, check its status using `systemctl`. The service can be started and enabled to start automatically after a reboot. If the service fails repeatedly, review its logs using `journalctl` and check the service's configuration files for errors.

```
sudo systemctl status service-name
```

```
sudo systemctl start service-name
```

```
sudo systemctl enable service-name
```

VI. Best Practices

A properly installed server should be maintained regularly to ensure security, reliability, and performance. The operating system and installed software should be updated regularly so that known vulnerabilities can be addressed.

Strong passwords should be used for all user accounts, and administrator accounts should not be shared between multiple users. Where possible, SSH key authentication should be used instead of password-based authentication. Unnecessary user accounts should be removed or disabled.

The server should use a firewall to restrict unauthorized network connections. Only ports and services required for the server's purpose should be exposed. Unnecessary software and services should also be removed or disabled.

Regular backups are essential for protecting important data and system configurations. Backups should be stored separately from the main server so that they remain available if the server experiences hardware failure, accidental deletion, or other problems. Backup restoration should also be tested periodically to ensure that the backups are usable.

System administrators should regularly monitor disk space, memory usage, CPU usage, network activity, running services, and system logs. Monitoring allows potential problems to be identified before they affect users or services.

Documentation should also be maintained whenever configuration changes are made. Important information such as IP addresses, hostnames, installed services, firewall rules, user accounts, backup procedures, and system changes should be recorded for future administrators.

Finally, administrators should follow the principle of least privilege. Users and services should only receive the permissions necessary to perform their assigned tasks. This reduces the potential impact of unauthorized access or compromised accounts.

VII. References

1. Canonical. *Ubuntu Documentation*.
<https://docs.ubuntu.com/>
2. Canonical. *Ubuntu Server Documentation*.
<https://ubuntu.com/server/docs/>
3. Canonical. *Ubuntu Security Documentation*.
<https://documentation.ubuntu.com/security/>
4. Canonical. *Ubuntu Security and Compliance Documentation*.
<https://documentation.ubuntu.com/security/compliance/>
5. Canonical. *Ubuntu Pro Client Documentation*.
<https://documentation.ubuntu.com/pro-client/en/latest/>